

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 2 – Case Study

Course Code:	DSA0307	Course Title:	Natural Language Processing
CO Assessed:	CO3 – Precision	Assessment:	Assessment Tool 2 – Case Study
Weightage:	40% of CIA		
CO5 Statement:	Apply N-gram models, smoothing techniques, part-of-speech tagging systems and to evaluate effective syntactic analysis. (BL-3)		
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Section / Batch:	BE-CSE-AI	Date:	07-08-26

Code of Conduct

*I **S.Muskan(192472182)** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: **S.Muskan**

Instructions

- Read all three case studies carefully before attempting the questions.
- Provide appropriate justifications and interpretations for every computed result.
- Use NLP concepts, algorithms, and terminology wherever applicable.
- Diagrams, flowcharts, tables, or mathematical formulations may be used to support your answers.
- Duration: 30 minutes.

Section / Topic	Focus	Case Study
N-Gram Language Models and Smoothing Techniques	Bigram and Trigram probability computation, Maximum Likelihood Estimation (MLE), Backoff models, Deleted Interpolation, Entropy calculation, uncertainty analysis, real-time next-word prediction	Case Study 1 – Smart Mobile Keyboard Prediction System
Part-of-Speech (POS) Tagging and Word Classes	Penn Treebank tagset, contextual ambiguity resolution, rule-based tagging, stochastic tagging (HMM), emission and transition probabilities, word class identification, chatbot language understanding	Case Study 2 – AI-Powered Customer Support Chatbot
Transformation-Based Tagging (Brill Tagger)	POS tag correction, transformation rules, contextual tagging, linguistic validation, tag sequence refinement, ambiguity handling	Case Study 3 – News Analytics and POS Tag Correction System
Corpus Statistics and Probabilistic Analysis	Word frequency counting, corpus-based probability estimation, entropy-based uncertainty measurement, tagging confidence evaluation	Case Study 3 – News Analytics and POS Tag Correction System
Integrated Syntactic Analysis and NLP Applications	Application of language models, POS tagging, probabilistic methods, corpus analysis, ambiguity resolution, and decision-making in real-world NLP systems	Case Studies 1, 2, and 3

CASE STUDY 1: Smart Mobile Keyboard Prediction System

A smartphone company is developing an AI-powered keyboard application that predicts the next word while users type emails, messages, and social media posts. The language model is trained on the corpus: *"Data science is powerful, data science drives innovation, data science is evolving."*

The development team employs Bigram and Trigram Language Models, Backoff Models, Deleted Interpolation, and Entropy Analysis to improve prediction accuracy and handle unseen word sequences. The following statistics are obtained from the corpus:

- $C(\text{data}) = 3$
- $C(\text{data science}) = 3$
- $C(\text{science is}) = 2$
- $C(\text{science drives}) = 1$
- $C(\text{science drives innovation}) = 1$

The interpolation weights are:

- $\lambda_1 = 0.5$ (Trigram)
- $\lambda_2 = 0.3$ (Bigram)
- $\lambda_3 = 0.2$ (Unigram)

The prediction probabilities after the word "science" are:

- $P(\text{is}) = 0.66$
- $P(\text{drives}) = 0.33$

Questions

1. Compute the Maximum Likelihood Estimate (MLE) of the Bigram Probability $P(\text{science} \mid \text{data})$. Interpret how this probability influences next-word prediction in a mobile keyboard application.
 2. A user types the unseen sequence "data science improves". Apply the Backoff Model and estimate the probability using lower-order n-grams. Explain why backoff is essential for real-time predictive text systems.
 3. Using Deleted Interpolation, calculate the probability of the sequence "data science is". Discuss how interpolation balances reliability and coverage in sparse datasets.
 4. Calculate the Entropy of the distribution $P(\text{is})=0.66$ and $P(\text{drives})=0.33$. Interpret the result and evaluate how entropy helps measure uncertainty and prediction confidence in intelligent keyboard systems.
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CASE STUDY 2: AI-Powered Customer Support Chatbot

A multinational airline deploys a customer support chatbot that processes user messages and automatically identifies the grammatical role of words before generating responses.

Consider the following user inputs:

1. "Book a flight ticket now."
2. "This book is interesting."

The chatbot uses the Penn Treebank POS Tagset and a Hidden Markov Model (HMM) for probabilistic tagging.

Given:

- $P(\text{book} \mid \text{VB}) = 0.6$
- $P(\text{book} \mid \text{NN}) = 0.4$
- $P(\text{Start} \rightarrow \text{VB}) = 0.5$
- $P(\text{Start} \rightarrow \text{NN}) = 0.5$

The system must determine whether the word "book" functions as a Verb (VB) or Noun (NN) depending on context.

Questions

1. Assign appropriate Penn Treebank POS tags to every word in both sentences. Justify the tag assigned to the word "book" using contextual clues.
 2. Using the HMM probabilities, compute the likelihood of "book" being tagged as a Verb (VB) in the sentence "Book a flight ticket now." Explain why the probabilistic model favors this tag.
 3. Compare Rule-Based Tagging and Stochastic (HMM) Tagging in resolving lexical ambiguity. Recommend which approach is more suitable for large-scale chatbot deployment and justify your answer.
 4. Evaluate the role of standardized POS tagsets and word classes in improving intent detection, response generation, and overall chatbot performance.
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CASE STUDY 3: News Analytics and POS Tag Correction System

A digital news analytics company processes millions of news articles daily. The platform uses a Transformation-Based Tagger (Brill Tagger) to improve tagging accuracy after an initial POS tagging phase.

The sentence processed by the system is:

"Economic growth increases employment."

Initial POS Tags:

- economic/JJ
- growth/NN
- increases/NNS
- employment/NN

A learned transformation rule states:

Change NNS to VBZ if the preceding word is tagged as NN.

The system also performs corpus-based frequency analysis and uncertainty evaluation.

Questions

1. Apply the transformation rule and update the POS tag sequence. Explain why the correction is linguistically appropriate.
2. Analyze the initial tagging errors and justify the corrected tag assignments using grammatical structure and sentence semantics.
3. Assuming the corpus contains the words:
 - economic (120 occurrences)
 - growth (450 occurrences)
 - increases (210 occurrences)
 - employment (380 occurrences)

Compute the word frequency distribution and explain how frequency information supports probabilistic POS tagging.

4. Evaluate how Transformation-Based Tagging improves tagging accuracy compared with the initial tagging stage. Discuss how entropy can be used to measure uncertainty in tag assignments before and after rule application.
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CASE STUDY 1 – SMART MOBILE KEYBOARD PREDICTION SYSTEM

Case Overview

The smartphone company is developing an AI-powered keyboard that predicts the next word while users type. The system uses **Bigram and Trigram Language Models, MLE, Backoff, Deleted Interpolation and Entropy Analysis**. The main challenge is to provide accurate predictions while also handling **unseen word sequences and uncertainty**.

Q1. Compute the MLE of the Bigram Probability $P(\text{science} \mid \text{data})$.

Case Understanding

The keyboard needs to determine how strongly "**science**" follows "**data**" in the training corpus. A Bigram Language Model considers two consecutive words.

Given

$C(\text{data})=3$ $C(\text{data science})=3$

Concept

Using Maximum Likelihood Estimation:

$P(w_i \mid w_{i-1}) = \frac{C(w_{i-1}w_i)}{C(w_{i-1})}$

Calculation

$P(\text{science} \mid \text{data}) = \frac{C(\text{data science})}{C(\text{data})} = \frac{3}{3} = 1.0$

Interpretation and Analysis

The probability is **1.0 or 100%**, meaning that every occurrence of "**data**" in the given corpus is followed by "**science**".

Therefore, when a user types:

data

the keyboard can confidently rank:

science

as the next-word prediction.

In a real keyboard, this can reduce typing effort and improve prediction accuracy. However, MLE alone cannot handle unseen sequences effectively.

Conclusion

MLE provides a strong prediction for frequently observed bigrams, but it should be combined with **backoff and smoothing** for real-world robustness.

Q2. Apply the Backoff Model to the unseen sequence "data science improves".

Case Understanding

The user enters:

"data science improves"

The trigram **"data science improves"** is unseen in the given corpus. Therefore, the system cannot directly use the highest-order trigram probability and must **back off to lower-order n-grams**.

Backoff Strategy

Trigram→Bigram→Unigram

The required trigram is:

$P(\text{improves} \mid \text{data, science})$

Since this trigram is unseen, we consider:

$P(\text{improves} \mid \text{science})$

The bigram **"science improves"** is also absent.

Finally, the unigram **"improves"** does not occur in the given corpus.

Therefore:

$P(\text{improves})=0$

Calculation

From Q1:

$P(\text{science} \mid \text{data})=1$

Hence, under the **unsmoothed probabilities provided in the case**:

$P(\text{data science improves}) \approx P(\text{science} \mid \text{data}) \times P(\text{improves}) = 1 \times 0 = 0$

Interpretation and Analysis

The zero probability demonstrates the **data sparsity problem**. A real keyboard will frequently encounter new words and word combinations that were not present in its training corpus.

Backoff is therefore essential because it allows the system to fall back from a higher-order model to a lower-order model instead of immediately failing.

However, in a practical system, **smoothing should accompany backoff** so that unseen words can receive a small non-zero probability.

Conclusion

Backoff + Smoothing = Better coverage for unseen sequences

This makes predictive text systems more robust in real-time use.

Q3. Using Deleted Interpolation, calculate the probability of "data science is".

Case Understanding

The keyboard must predict "is" after the context "data science". Instead of relying only on the trigram, Deleted Interpolation combines evidence from **trigram, bigram and unigram models**.

Given

$$\lambda_1 = 0.5, \lambda_2 = 0.3, \lambda_3 = 0.2$$

where:

- λ_1 = Trigram weight
- λ_2 = Bigram weight
- λ_3 = Unigram weight

Step 1 – Trigram Probability

The phrase "data science is" occurs twice.

$$P(\text{is} \mid \text{data, science}) = \frac{C(\text{data science is})}{C(\text{data science})} = \frac{2}{3} = 0.6667$$

Step 2 – Bigram Probability

$$P(\text{is} \mid \text{science}) = \frac{C(\text{science is})}{C(\text{science})}$$

Since "science" occurs 3 times:

$$P(\text{is} \mid \text{science}) = \frac{1}{3} = 0.3333$$

Step 3 – Unigram Probability

The corpus contains 12 words and **is** occurs twice.

$$P(\text{is}) = \frac{2}{12} = 0.1667$$

Step 4 – Deleted Interpolation

$$\begin{aligned} \text{PDI}(\text{is} \mid \text{data, science}) &= \lambda_1 P(\text{is} \mid \text{data, science}) + \lambda_2 P(\text{is} \mid \text{science}) + \lambda_3 P(\text{is}) \\ &= (0.5)(0.6667) + (0.3)(0.3333) + (0.2)(0.1667) \\ &= 0.3333 + 0.1000 + 0.0333 = 0.4667 \end{aligned}$$

If the examiner interprets "**probability of the sequence**" as the complete sequence probability:

$$P(\text{data}) = \frac{1}{4} = 0.25$$

$$\begin{aligned} \text{Therefore: } P(\text{data science is}) &= P(\text{data})P(\text{science} \mid \text{data})\text{PDI}(\text{is} \mid \text{data, science}) \\ &= 0.25(1)(0.4667) \\ P(\text{data science is}) &\approx 0.1167 \end{aligned}$$

Interpretation and Analysis

The interpolation gives different weights to different levels of context.

Model Probability Weight

Trigram	0.6667	0.5
Bigram	0.3333	0.3
Unigram	0.1667	0.2

The trigram receives the highest weight because it provides the most specific context. The lower-order models provide additional coverage.

Conclusion

Deleted Interpolation balances **contextual reliability and coverage**, making it useful when training data is sparse.

Q4. Calculate the Entropy of $P(\text{is})=0.66$ and $P(\text{drives})=0.33$.

Case Understanding

The keyboard has two possible next-word predictions after "**science**". Entropy measures how uncertain the keyboard is about which word will occur next.

Formula

$$H(X) = -\sum P(x) \log_2 P(x)$$

Calculation

$$H(X) = -[0.66 \log_2 (0.66) + 0.33 \log_2 (0.33)] \quad H(X) \approx 0.923 \text{ bits}$$

The supplied probabilities total **0.99** because they are rounded. If normalized, entropy is approximately **0.918 bits**.

Interpretation

The prediction "**is**" has a much higher probability than "**drives**". Hence, the keyboard has relatively low uncertainty and can prioritize "**is**".

Analysis

Entropy can help the keyboard measure prediction confidence:

- **Low entropy** → **high confidence**
- **High entropy** → **high uncertainty**

If entropy is high, the keyboard can display several suggestions instead of strongly favoring one word.

Conclusion

Entropy provides a quantitative measure of uncertainty and can support intelligent decision-making in next-word prediction.

CASE STUDY 2 – AI-POWERED CUSTOMER SUPPORT CHATBOT

Case Overview

The airline chatbot must understand user messages and identify the grammatical role of each word before generating responses. The main challenge is **lexical ambiguity**, particularly the word "**book**", which can function as either a noun or a verb. The system uses the **Penn Treebank POS Tagset** and **HMM-based stochastic tagging**.

Q1. Assign Penn Treebank POS tags to both sentences.

Sentence 1

"Book a flight ticket now."

Word	Tag	Reason
Book	VB	Action/request
a	DT	Determiner
flight	NN	Singular noun
ticket	NN	Singular noun
now	RB	Adverb

Tagged sentence

Book/VB a/DT flight/NN ticket/NN now/RB

Case Analysis

Here, **book** is an action performed by the user, so it is a **verb (VB)**.

Sentence 2

"This book is interesting."

Word	Tag	Reason
This	DT	Determiner
book	NN	Singular noun
is	VBZ	Present-tense verb
interesting	JJ	Adjective

Tagged sentence

This/DT book/NN is/VBZ interesting/JJ

Case Analysis

Here, **book** refers to an object and acts as the subject, so it is a **noun (NN)**.

Conclusion

The word **book** is contextually ambiguous:

book→VB or NN

Its correct POS tag depends on the surrounding words and sentence structure.

Q2. Using HMM probabilities, compute the likelihood of "book" being tagged VB.

Given

$$P(\text{book} \mid \text{VB})=0.6 \quad P(\text{book} \mid \text{NN})=0.4 \quad P(\text{Start} \rightarrow \text{VB})=0.5 \quad P(\text{Start} \rightarrow \text{NN})=0.5$$

HMM Concept

An HMM combines:

- **Transition probability**
- **Emission probability**

For VB:

$$P(\text{VB}, \text{book}) = P(\text{Start} \rightarrow \text{VB}) P(\text{book} \mid \text{VB})$$

Calculation

$$=(0.5)(0.6) \quad P(\text{VB}, \text{book})=0.30$$

For comparison:

$$P(\text{NN}, \text{book}) = (0.5)(0.4) = 0.20$$

Since:

$$0.30 > 0.20$$

the model favors **VB**.

Normalized:

$$P(\text{VB} \mid \text{book}) = \frac{0.30}{0.30 + 0.20} = 0.60 \quad P(\text{VB} \mid \text{book})=0.60$$

Interpretation

The HMM favors the verb interpretation because **"book" is more strongly associated with VB than NN according to the given emission probabilities.**

Case Analysis

For:

Book a flight ticket now

the VB interpretation is also linguistically supported because the sentence begins with an action/request.

Conclusion

The HMM combines statistical evidence with contextual information to resolve lexical ambiguity.

Q3. Compare Rule-Based and Stochastic HMM Tagging.

Case Understanding

The chatbot must distinguish words such as "**book**" that can have multiple grammatical roles. Two approaches are possible.

Feature	Rule-Based Tagging	HMM/Stochastic Tagging
Method	Hand-crafted rules	Probability-based
Context	Predefined linguistic rules	Statistical context
Ambiguity	Depends on available rules	Uses probabilities
Scalability	Difficult for many rules	More scalable
Maintenance	Manual updates required	Can retrain using data
Flexibility	Limited	Better for varied language

Analysis

Rule-based tagging can perform well when linguistic rules are clearly defined, but large chatbots encounter many ambiguous words and sentence structures.

HMM tagging can estimate the most probable tag using observed language patterns.

Recommendation

For large-scale chatbot deployment:

Stochastic/HMM Tagging is more suitable

because it is more adaptable to large amounts of language data and can statistically resolve ambiguity.

A practical system can further improve reliability by combining **statistical tagging with linguistic rules**.

Conclusion

For a large-scale chatbot, a **probabilistic approach or hybrid approach** provides better scalability and ambiguity handling than a purely rule-based system.

Q4. Evaluate the role of standardized POS tagsets and word classes.

Case Understanding

A chatbot needs a consistent representation of grammatical information before performing tasks such as intent detection and response generation.

Role of POS Tagsets

The Penn Treebank tagset provides standardized labels such as:

NN, VB, JJ, RB, DT, VBZ

Application in Chatbot

1. Intent Detection

In:

Book a flight.

book/VB indicates an action/request.

2. Ambiguity Resolution

The system can distinguish:

book/NN

from:

book/VB

based on context.

3. Response Generation

Correct grammatical information helps the chatbot construct appropriate responses.

4. Syntactic Understanding

POS tags provide information about sentence structure and word relationships.

Overall Analysis

The processing pipeline can be represented as:

POS Tagging→Syntactic Analysis→Intent Detection→Response Generation

Conclusion

Standardized POS tagsets improve **consistency, ambiguity resolution, intent detection, syntactic understanding and response quality**, thereby improving overall chatbot performance.

CASE STUDY 3 – NEWS ANALYTICS AND POS TAG CORRECTION SYSTEM

Case Overview

The news analytics company processes millions of articles and uses a **Transformation-Based Tagger (Brill Tagger)** to correct errors made during initial POS tagging. The system also uses corpus frequency and entropy to evaluate tagging confidence.

Q1. Apply the transformation rule.

Initial Tags

economic/JJ growth/NN increases/NNS employment/NN

Given Rule

Change **NNS** → **VBZ** if the preceding word is tagged as **NN**.

Rule Application

The preceding word of **increases** is:

growth/NN

Therefore, the condition is satisfied:

NNS→VBZ

Corrected Sequence

economic/JJ growth/NN increases/VBZ employment/NN

Linguistic Analysis

The sentence is:

Economic growth increases employment.

"Growth" is a singular subject, so the verb must agree with it:

growth→increases

Thus, **increases** functions as a third-person singular present verb (**VBZ**), not a plural noun (**NNS**).

Conclusion

The Brill transformation correctly uses contextual information to repair the initial POS tagging error.

Q2. Analyze the initial tagging errors.

Initial Sequence

economic/JJ growth/NN increases/NNS employment/NN

The main error is:

increases/NNS

Corrected Tag

increases/VBZ

Linguistic Analysis

Word	Correct Tag	Reason
economic	JJ	Modifies "growth"
growth	NN	Singular subject
increases	VBZ	Action performed by singular subject
employment	NN	Object of the verb

The sentence follows:

Subject+Verb+Object growth+increases+employment

Therefore, the transformation is grammatically and semantically justified.

Conclusion

The Brill Tagger improves the initial output by using **contextual grammatical evidence** rather than considering the word in isolation.

Q3. Compute the word frequency distribution.

Given

Word	Frequency
economic	120
growth	450
increases	210
employment	380

Total Corpus Frequency

$N=120+450+210+380$ $N=1160$

Probability Formula

$P(w)=\frac{NC(w)}{N}$

Frequency Distribution

Word	Count	Probability	Percentage
economic	120	0.1034	10.34%
growth	450	0.3879	38.79%
increases	210	0.1810	18.10%
employment	380	0.3276	32.76%
Total	1160	1.0000	100%

Analysis

The most frequent word is:

growth

with probability:

$P(\text{growth})=0.3879$

Frequency information provides statistical evidence for corpus-based probability estimation and can contribute to probabilistic POS tagging.

However, **frequency alone cannot determine grammatical role**. Context is still required to distinguish different POS interpretations.

Conclusion

Corpus statistics provide useful probabilistic evidence, but they should be combined with **contextual and syntactic information** for reliable tagging.

Q4. Evaluate Transformation-Based Tagging and Entropy. Case Understanding

The initial POS tagger incorrectly identifies:

increases/NNS

The Brill Tagger applies a learned contextual rule to correct it.

Transformation

growth/NN + increases/NNS

satisfies:

Change NNS to VBZ when the preceding word is NN.

Therefore:

increases/NNS → increases/VBZ

Before Transformation

economic/JJ growth/NN increases/NNS employment/NN

After Transformation

economic/JJ growth/NN increases/VBZ employment/NN

Analysis

Transformation-Based Tagging improves accuracy because it identifies errors using **contextual linguistic rules**.

Instead of simply asking:

"What tag does the word 'increases' usually have?"

the system considers:

"What tag is appropriate when 'increases' follows a singular noun such as 'growth'?"

This provides more linguistically meaningful correction.

Entropy Analysis

Entropy measures uncertainty:

$$H(X) = -i \sum P(x_i) \log_2 P(x_i)$$

Before correction, the system may have uncertainty between competing tags such as:

NNSandVBZ

After applying the correct transformation, confidence becomes concentrated on:

VBZ

Therefore:

Before correction: Higher uncertainty ↓ Brill transformation ↓ After correction: Lower uncertainty

Important Justification

The case study **does not provide probabilities for NNS and VBZ**, so an exact numerical entropy value cannot be calculated. We can only conclude qualitatively that uncertainty is reduced after a successful correction.

Conclusion

Transformation-Based Tagging improves the initial tagging stage by applying **context-sensitive linguistic corrections**, while entropy can be used to evaluate whether uncertainty in POS assignment decreases after correction.

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand

Q. No. / Criteria Level	Understanding of Case	Application of Concepts	Analysis	Solution Design	Clarity	Sub. Total	Total %
1							
2							
3							

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____